Engine Model(s): PT6A

INTRODUCTION

1. General

A. This maintenance manual contains the recommended procedures for maintaining PT6A-21/-27/-28 engines manufactured by Pratt & Whitney Canada. Personnel involved with engine maintenance should acquaint themselves with the contents of the following introductory paragraphs for full comprehension of the information contained in this manual.

B. Any discrepancies, problems or suggestions regarding this publication should be forwarded in writing, using the Pratt & Whitney Canada Customer Feedback Sheet (RSVP), to:

Pratt & Whitney Canada Corp.
1000 Marie-Victorin Blvd
Longueuil, Quebec
Canada
J4G 1A1

Attention: Manager Publications Dept (05MS1)

Website: www.pwc.ca
Email: customer.service@pwc.ca

C. Customer Feedback Sheets are enclosed with new manuals and each subsequent revision. Additional forms may be obtained by contacting: The Supervisor, Publications Customer Services, at the above address.

D. The on-line Customer Feedback Sheet (RSVP form) is also available on the Pratt & Whitney Canada website (www.pwc.ca).

E. Procedures in this manual may be accomplished with the engine installed or removed from the airframe. This manual will be revised as necessary to incorporate any changes as they arise and also incorporate latest approved procedures and data. This manual is not customized to any particular airframe installation. Configurational variations that may arise as a result of installation or peculiar requirement will be dealt with in the text as they occur.

2. Customer Service

Customer Service representatives maintain contact with operators and service activities and are available for investigation of any specific difficulty or problem. Request for assistance should be directed to:

Pratt & Whitney Canada Corp.
1000 Marie-Victorin Blvd.
Longueuil, Quebec
Canada
J4G 1A1

Attention: Customer Service

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Request for AOG support may also be directed to:

Pratt & Whitney Canada Corp.
1000 Marie-Victorin Blvd.
Longueuil, Quebec
Canada
J4G 1A1

Attention: Customer First Centre (CFirst) (01TS4)

REF: CFirst (24-hour service)

US and Canada: 1-800-268-8000

International: (International Access Code) -8000-268-8000

Other: 1-450-647-8000

Fax: 1-450-647-2888

Email: CFirst@pwc.ca

Telephone or write to the manager of the Spare Parts Sales Department to get details about how to order parts. You can also get information from the Customer First Centre (CFirst).

Pratt & Whitney Canada Corp.
1000 Marie-Victorin Blvd.
Longueuil, Quebec
Canada
J4G 1A1

Attention: Spare Parts Sales Department

3. Accessories Support

Accessories can be sent for repair or full overhaul to:

P&WC Accessories Services
7007, De La Savane,
St-Hubert, Quebec
Canada
J3Y 8Y9

4. How to Use this Manual

A. Chapter/Section/Subject Numbering

Subject matter in this manual is separated into specific Chapters, Sections and Subjects in accordance with ATA Specification No. 100. Each chapter is divided into sections and each section is divided into units. These three elements are shown in large bold numbers at the lower right hand corner of each page, where applicable. Chapters are separated by tab dividers.

Chapters contained in this manual are as follows:

AIRWORTHINESS LIMITATIONS	Life Limited Parts
61	Propeller Control
70	Standard Practices
71	Power Plant
72	Engine
73	Fuel and Control Systems
74	Ignition System
75	Air System
76	Engine Controls
77	Engine Indicating System
79	Oil System

B. Chapter 72

The engine chapter is further broken down into sections:

72-01 Power Section

72-10 Reduction Gearbox

NOTE: An engine that has more than one RGB, will use sequential numbers i.e., 72-11-00, 72-12 -00 etc.

72-20 Air Inlet Section

72-30 Compressor Section

72-40 Combustion Section

72-50 Turbine Section

72-60 Accessory Gearbox

C. Other Chapters

Chapter/Section breakdown for the other Chapters is extensive and is shown in the Table of Contents for each Chapter.

5. Engine Part Nomenclature

For engine areas, bearings, stations, and flanges, refer to 72-00-00, Description and Operation, Figures 1 through 4.

6. Directional References

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The terms right and left, clockwise and counterclockwise, upper and lower and similar references will apply to the engine as viewed from the rear, with the engine in a horizontal position and the accessory gearbox facing the viewer.

7. Maintenance Concept

The maintenance functions detailed in this manual fall into two categories: Line Maintenance and Heavy Maintenance. Repairs beyond the levels detailed in this manual are not recommended as maintenance functions and should be done by an approved overhaul facility.

A. Line Maintenance

The scope of line maintenance consists of removal and installation of external components and engine accessories. All procedures in this manual are to be considered line maintenance in the absence of the words "(Heavy Maintenance Only)" appearing with the procedure title in Maintenance Practices.

B. Heavy Maintenance

Heavy maintenance details removal, installation and repair of components or parts normally considered beyond the capabilities of the average line maintenance facility.

8. Supplementary Publications

Personnel concerned with the maintenance of the engines should familiarize themselves with the contents of the Illustrated Parts Catalog (P&WC Manual P/N 3013244) which lists and describes the saleable parts of the engines and illustrates their inter-relationships.

9. Service Bulletins

Service bulletins will be issued as required to provide information or instructions for modifying engines or parts to the latest configuration.

10. Abbreviations and Symbols

The following is a list of abbreviations and symbols used throughout this manual:

AGB	Accessory Gearbox
AMS	Aerospace Material Specification
Assy	Assembly
ATA	Air Transport Association
beta	Range of propeller blade angle between fine pitch stop and maximum reverse pitch setting
BRG	Bearing
CCW	Counterclockwise
CG	Center of Gravity

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CSU	Constant Speed Unit
CT	Compressor Turbine
CW	Clockwise
°C	Degrees Celsius (Centigrade)
°F	degrees Fahrenheit
°R	degrees Rankin
°K	degrees Kelvin
del	delta (P observed/P Standard Day)
del P	Engine Torquemeter Pressure, in. Hg.
del T5	T5 Temperature Trim Value
DELRTH	Delta Root Theta
Dia.	Diameter
Dim	Dimension
eshp	equivalent shaft horse power
FCU	Fuel Control Unit
FI	Flight Idle
FIR	Full Indicator Reading
FOD	Foreign Object Damage
FPI	Fluorescent Penetrant Inspection
FWD (Fwd)	Forward
G	Gravitational Constant
GGM	Gas Generator Module
GI	Ground Idle
Hg	Mercury
HP	High Pressure
hr	hour
HSI	Hot Section Inspection
ID	Inside Diameter
Imp	Imperial

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in.	inch
in.Hg.	Inches of Mercury
IPC	Illustrated Parts Catalog
ITT (T5)	Interturbine Temperature
kg	kilogram
lb.	pound
lb.in./lbf.in.	pound force inch
LH	Left-hand
LP	Low Pressure
Max.	Maximum
min	minute
Min.	Minimum
mm	millimeter
NA	Not Applicable
Nf (NF)	Power Turbine Speed (rpm or %)
Ng (NG)	Gas Generator Speed (rpm or %)
No.	Number
Np	Propeller Speed (rpm or %)
OD	Outside Diameter
Pam	Ambient Barometric Pressure
Para.	Paragraph
PD	Pitch Diameter
P/N	Part Number
PP	Plenum Pressure
pph	pounds per hour
Psgb	Reduction Gearbox Static Pressure
psi	pounds per square inch
psig	pounds per square inch gage

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PSM	Power Section Module
PT	Power Turbine
PTO	Power Take Off
P&WC	Pratt & Whitney Canada Corp.
PWC	Special Tool Prefix Letters - Engine
Px	Control Pressure (Bleed Valve)
Py	Governor Servo Pressure
Po	Bypass Pressure
P1	Unmetered Fuel Pressure or Intake Pressure
P2	Metered Fuel Pressure
P3	Compressor Discharge Air Pressure
(REF.)	reference
RGB	Reduction Gearbox (Propeller)
RH	Right-hand
RTD	Resistance Temperature Device
RTH	Root Theta
rpm	revolutions per minute
SB	Service Bulletin
SCFM	Standard Cubic Feet per Minute
SED	Small Exit Duct
sfc	specific fuel consumption (lb/shp/hr)
SG	Specific Gravity
shp	shaft horse power
SOI	Specific Operating Instructions
SPOP	Service Process Operation Procedure
T	Temperature
tam	Ambient Air Temperature
TBA	To be advised
TBO	Time Between Overhaul

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theta	T observed/T standard day
TIG	Tungsten Inert Gas (Welding method)
T1	Engine Inlet Temperature
T5 (ITT)	Inter-turbine Temperature
UOS	Unless Otherwise Specified
USA	United States of America
Wf	Fuel Flow (lb/hr)

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